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RESEARCH ARTICLE

Adaptive Focal Loss for Keypoint-Based Deep Learning Detectors Addressing Class Imbalance

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ABSTRACT Keypoint-based deep learning detectors have proven highly effective in object detection tasks by predicting specific keypoints to determine object classification and location. Examples include CornerNet, CenterNet, ExtremeNet, RetinaNet, FCOS, and ObjectBox. Despite their strengths, these methods are particularly susceptible to class imbalance, which can result in poor detection performance for less frequent classes. Popular solutions such as hard sampling, soft sampling, and sampling-free methods have been proposed to tackle this issue. However, these approaches often have inherent limitations, including sensitivity to hyperparameters and neglecting the gradient dynamics of the loss function. To address these challenges, this paper proposes Adaptive Focal Loss (AFL), which combines the strengths of Focal Loss (FL) and Class-Balanced Loss (CBL) while introducing an Adaptive Gradient Function. This function is specifically designed to mitigate the impact of large gradients during the early stages of training. Extensive experiments demonstrate that AFL significantly outperforms classical sampling methods across key metrics on the MS COCO, LVIS, and PASCAL VOC datasets. Notably, on the LVIS dataset, AFL achieves an average improvement of over 3% AP_r compared to the baseline Focal Loss across multiple keypoint-based detectors, showcasing its effectiveness in enhancing rare class accuracy. Furthermore, AFL delivers substantial improvements in the $AP_{0.5\sim0.95}$ metric, surpassing the baseline Focal Loss by an average of over 2%, 1%, and 0.2% on the LVIS, MS COCO, and PASCAL VOC datasets, respectively. This highlights AFL's capability to enhance the overall accuracy of foreground objects. By providing a more balanced and robust solution to class imbalance, AFL demonstrates superior performance in challenging scenarios, making it a valuable advancement in keypoint-based detection.

INDEX TERMS Deep learning algorithm, object detection, keypoint-based deep learning detector, class imbalance, sampling method.

I. INTRODUCTION

Keypoint-based deep learning detectors are a class of object detection models that rely on identifying specific keypoints, such as corners, centers, or other significant points of objects, to determine their presence and location. As illustrated in Figure 1, models such as CornerNet [1], CenterNet [2], ExtremeNet [3], RetinaNet [4], FCOS [5], and ObjectBox [6] aim to predict the precise locations of these keypoints and use them to infer object boundaries, poses, or other relevant features. By focusing on keypoints, these models simplify the detection process, improve accuracy, and support various applications such as human pose estimation, object localization, and multi-person tracking [7]. They are particularly effective for tasks where the spatial configuration of an object's parts plays a critical role in its identification and classification [8].

However, keypoint-based detectors often encounter more severe challenges with class imbalance than region proposal-based detectors. Since they directly predict keypoints across the entire image, there is a significant imbalance between the numerous background points and the relatively few foreground keypoints [9], [10]. This imbalance can bias the model toward predicting background, thereby reducing its

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FIGURE 1. **Illustration of different keypoint-based deep learning detectors.**

ability to accurately detect and classify objects, especially for less frequent classes [11], [12]. In contrast, region proposal-based detectors generate a limited number of candidate regions that are more likely to contain objects. This mechanism helps mitigate class imbalance by narrowing the model's focus to potential object locations rather than the overwhelming majority of background pixels [13], [14].

For proposal-based deep learning detectors, the gap between positive and negative samples is substantial. Typically, both proposal-based and keypoint-based detectors divide the image into $k \times k$ grids, as depicted in Figure 2. Proposal-based detectors generate b candidate bounding boxes in each grid, while keypoint-based detectors predict a centerpoint or keypoint in each grid. For example, $k \times k$ grids generate $k \times k \times b$ bounding boxes, of which only $\frac{1}{k \times k \times b}$ represents the most suitable sample. In contrast, for detectors based on keypoints or centerpoints, the most suitable sample in $k \times k$ grids is $\frac{1}{k \times k}$, significantly reducing the positive-to-negative sample ratio. However, most proposal-based detectors leverage a two-stage classification approach to address the class imbalance problem. These detectors first classify background and foreground samples and then classify foreground categories. Keypoint-based detectors, lacking proposals, cannot employ such methods. According to the paper from Chen et al. [15], the foreground-to-background ratio in object detection tasks is approximately 100:100k. In region proposal-based methods, the initial stage filters out foreground regions, reducing the classification task to identifying positive samples from around 100 foreground candidates. In contrast, keypoint-based detectors must directly classify positive samples from approximately 100k/b candidate points, where b typically equals around 10. This means the classification must be performed among roughly 10k candidate points, significantly increasing the computational complexity and the challenge of identifying positive samples accurately. Consequently, keypoint-based detectors are more prone to suffering from class imbalance [16], [17], [18].

To address class imbalance in deep learning, strategies such as hard sampling methods, soft sampling methods, and sampling-free methods are commonly employed. These approaches, including hard sampling techniques like the Overlap Sampler [19] and Single-Shot Refinement Network [20], soft sampling methods such as Equalization Loss [21] and Generalized Focal Loss [22], and sampling-free approaches like Residual Objectness [23], aim to enhance model performance on underrepresented classes by modifying the data distribution, adjusting the loss function, or dynamically tuning model parameters. However, these methods often fail to fully address challenges related to the gradient dynamics of the loss function, especially under extreme class imbalance, which can result in suboptimal updates during training [24]. Additionally, a common limitation among these methods is the introduction of increased computational complexity and hyperparameter tuning difficulties, which can hinder model generalization, particularly when applied to highly diverse or large-scale datasets [25].

In response to these challenges, this paper proposes Adaptive Focal Loss (AFL), which builds upon the foundation of Focal Loss (FL) and draws inspiration from Class-Wise Difficulty-Balanced Loss [26] and Class-Balanced Loss (CBL) [27]. Through rigorous mathematical analysis, this paper identifies that class imbalance often leads to

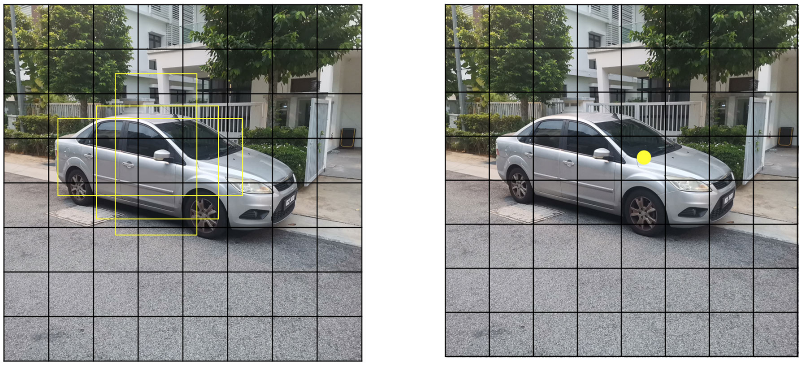

FIGURE 2. Comparison between candidate bounding boxes of the region proposal-based detector and candidate keypoints of the keypoint-based detector in an image.

excessively large gradients during the initial training stages, thereby impeding effective learning [23], [28]. To mitigate this issue, an Adaptive Gradient Function is introduced as an activation function to reduce the gradient impact during early training. The proposed AFL employs FL for foreground and background classification while incorporating CBL principles to balance the learning process across diverse sample difficulties and class distributions.

AFL has demonstrated significant performance improvements across multiple datasets, including MS COCO 2017 [29], LVIS v5.0 [30], and PASCAL VOC 2007 [31]. In particular, on the LVIS dataset, AFL achieves an average improvement of over 3% in the AP_r metric compared to the baseline Focal Loss across various keypoint-based detectors, underscoring its effectiveness in improving rare class accuracy. Additionally, AFL delivers consistent gains in the $AP_{0.5\sim0.95}$ metric, surpassing the baseline Focal Loss by an average of over 2%, 1%, and 0.2% on the LVIS, MS COCO, and PASCAL VOC datasets, respectively. These results highlight AFL's ability to enhance not only the detection of rare classes but also the overall accuracy of foreground objects across diverse datasets.

Our contributions can be summarized in three key aspects: (1) identifying the negative impact of excessively large gradients during early training stages in keypoint-based detectors and proposing an Adaptive Gradient Function to address this issue; (2) designing AFL, which effectively combines the strengths of FL and CBL to tackle class imbalance; and (3) demonstrating AFL's superior performance across multiple datasets, including MS COCO, PASCAL VOC, and particularly the long-tailed LVIS dataset, where it significantly improved rare-class object detection.

II. RELATED WORKS

A. DEEP LEARNING DETECTORS

Deep learning detectors are broadly categorized into two main types: region proposal-based detectors and keypoint-based detectors. Region proposal-based detectors, such as Faster R-CNN [32], generate a set of candidate regions (proposals) that are likely to contain objects, and then classify and refine these regions to detect objects. This two-stage approach provides high accuracy, particularly in challenging scenarios. On the other hand, keypoint-based detectors directly predict the locations of specific points, or keypoints, associated with objects, using these to infer object boundaries, poses, or other relevant features. For example, CornerNet [1] detects objects by identifying their top-left and bottom-right corners, while CenterNet [2] improves on this by detecting the center point along with the corners for more accurate localization. ExtremeNet [3] takes a similar approach but uses the extreme points (top, bottom, left, right) to define object boundaries. RetinaNet [4] and FCOS [5] are keypoint-based detectors that predict keypoints densely across the image, with FCOS using center points to define bounding boxes and RetinaNet introducing FL to handle class imbalance. ObjectBox [6] is another keypoint-based method that detects objects by predicting the four corner points and the center point, using these to directly generate bounding boxes. These methods streamline the detection process and are particularly well-suited for tasks requiring high-speed processing and flexibility in object localization [33].

B. SAMPLING METHODS ADDRESSING CLASS IMBALANCE

Class imbalance is a significant challenge in machine learning, particularly in object detection and classification tasks where minority classes are underrepresented [34], [35]. This imbalance can lead to biased models that underperform on minority classes, necessitating the development of specialized methods to mitigate this issue. Various strategies have been proposed, broadly categorized into hard sampling, soft sampling, and sampling-free approaches. Each category offers unique techniques to address the skewness in data distribution, aiming to improve model performance across all classes [36], [37].

Hard sampling methods directly modify the training data distribution to balance class representation. The Overlap Sampler for region-based object detection, introduced by

Chen et al. (2020), addresses class imbalance by selectively sampling overlapping regions to reduce bias in object detection tasks [19]. Similarly, Nie et al. proposed the Enriched Feature Guided Refinement Network, which enhances object detection by focusing on feature refinement for minority classes, thereby improving detection accuracy in imbalanced datasets [38]. Another significant contribution is the Single-Shot Refinement Neural Network by Zhang et al., which iteratively refines object candidates in a single-shot manner, effectively addressing the imbalance between foreground and background classes [20].

Soft sampling methods focus on adjusting the learning process rather than the data itself. Lin et al. introduced FL, a widely used technique in object detection that addresses the imbalance between foreground and background classes by dynamically down-weighting the loss contribution of easy examples, allowing the model to focus more on hard, misclassified examples [4]. This approach is particularly effective in scenarios with extreme class imbalance, such as dense object detection. Cui et al. proposed CBL, which adjusts the loss based on the effective number of samples, giving more weight to underrepresented classes to balance the learning process across all categories [27]. Tan et al. introduced Equalization Loss, a technique designed for long-tailed object recognition that adjusts the loss function to give more weight to minority classes, thereby improving recognition accuracy [21]. Li et al. proposed Generalized Focal Loss, which enhances dense object detection by learning distributed and qualified bounding boxes, addressing imbalance by focusing more on difficult-to-classify instances [22]. Additionally, Cao et al. developed Prime Sample Attention, a mechanism that prioritizes key samples during object detection, ensuring that minority class instances receive adequate attention during training [11].

Sampling-free methods bypass traditional data sampling and focus on dynamically adjusting model parameters during training. Chen et al. questioned the necessity of heuristic sampling in deep object detectors, proposing a sampling-free mechanism that adjusts classification gradients and loss scaling to handle class imbalance effectively [13]. In related work, Chen et al. introduced Residual Objectness, a method that reduces imbalance by incorporating residual connections into the model, enhancing performance on minority classes without altering the data distribution [23]. Oksuz et al. further contributed with Rank & Sort Loss, a technique for object detection and instance segmentation that ranks and sorts predictions to optimize performance in imbalanced scenarios [33].

In summary, the literature on class imbalance reveals a diverse array of methods aimed at improving the model performance of minority classes. Hard sampling techniques modify the data distribution, soft sampling adjusts the learning process, and sampling-free methods optimize model parameters dynamically. Each approach offers valuable insights and tools for mitigating the challenges posed by class imbalance in machine learning.

III. METHOD

A. INVESTIGATION OF CLASS IMBALANCE ON THE TRAINING OF KEYPOINT-BASED DETECTORS

To explore the influence of class imbalance on the training of keypoint-based detectors, this paper investigates the reasons using mathematical principles. This paper reveals that the observed decrease in performance can be attributed to the disproportionate gradient magnitude [39]. This paper first mathematically introduces foreground-background imbalance and sampling methods, then extends the imbalance to class imbalance in keypoint-based networks between positives and negatives. Then, this paper analyzes their impact on the training process theoretically. Finally, the experimental results demonstrate that the gradient magnitude is the key factor affecting the detector accuracy [40].

1) LOSS FUNCTION OF DEEP LEARNING DETECTORS

Because training is based on the loss function, this paper examines class imbalance from the perspective of the loss function. The general loss function of a deep learning detector is expressed as Equation 1

$$L_{total} = \lambda_{cls} L_{cls} + \lambda_{loc} L_{loc} \tag{1}$$

Here, λ_{cls} and λ_{loc} are weighting coefficients that balance the relative importance of classification and regression losses. L_{cls} is responsible for object classification, while L_{loc} handles object location and bounding box size. L_{cls} is usually the Cross-Entropy Loss [41] and its variants, such as CBL [27], and FL [4].

2) CROSS ENTROPY LOSS

In deep learning algorithms, especially in classification and object detection tasks, the Cross-Entropy Loss function is widely used to measure the difference between the probability distribution predicted by the model and the probability distribution of the true labels [41]. This loss function is crucial for object detectors as it helps the model learn to classify different objects accurately.

$$CE(p, y) = \begin{cases} -\log(p) & \text{if} \quad y = 1, \\ -\log(1-p) & \text{otherwise.} \end{cases} \tag{2}$$

In Equation 2, p represents the predicted probability and y represents the true label. This paper introduces a new concept p_t, and Equation 2 can be rewritten as Equation 3

$$\begin{aligned} CE(p, y) &= CE(p_t) = -\log(p_t), \\ p_t &= \begin{cases} p & \text{if} \quad y = 1, \\ 1-p & \text{otherwise.} \end{cases} \end{aligned} \tag{3}$$

Cross-entropy Loss performs well in multi-class classification problems because it directly optimizes the classification probability. Additionally, when the prediction deviates from the true label, Cross-Entropy Loss generates a larger gradient, which helps speed up learning and improve model convergence. However, this loss function has limitations [27].

For example, in cases of extreme class imbalance, Cross-Entropy Loss may cause the model to prefer frequent classes and ignore rare ones. To overcome this class imbalance problem, sample resampling techniques or modifications to the loss function, such as introducing class weights, may be necessary [42].

3) OPTIMIZATION OF LOSS FUNCTION

First, the loss function L is defined as Equation 4

$$L = -s\sum_{i}^{N}[y_i log(p_i) + (1 - y_i)log((1 - p_i))] \tag{4}$$

where s is a scaling factor, y_i is the label of the i-th sample, and p_i is the probability that the i-th sample is predicted to be a positive class.

The gradient descent optimization formula [13] is Equation 5

$$\theta^{t+1} = \theta^t - \eta\frac{\partial L}{\partial\theta^t} \tag{5}$$

Here, θ^t represents the parameters after t iterations, and η is the learning rate. At the beginning, $t = 0$ means the model has not been iterated, So this paper assumes $\theta^t = \theta^0$ before training.

The gradient of the loss function with respect to the parameters is calculated as Equation 6

$$\begin{aligned}\frac{\partial L}{\partial\theta^0} &= -s\sum_{i}^{N}\left[y_i\frac{\partial log(p_i)}{\partial\theta^0} + (1 - y_i)\frac{\partial log((1 - p_i))}{\partial\theta^0}\right]\\ &= -s\sum_{i}^{N}\left[y_i\cdot\frac{1}{p_i}\cdot\frac{\partial p_i}{\partial\theta^0} - (1 - y_i)\cdot\frac{1}{1 - p_i}\cdot\frac{\partial p_i}{\partial\theta^0}\right]\end{aligned} \tag{6}$$

The gradient norm is calculated as Equation 7

$$\left\|\frac{\partial L}{\partial\theta^0}\right\| = s\left|N^f\frac{1}{p} - N^b\frac{1}{1 - p}\right|\cdot\left\|\frac{\partial p}{\partial\theta^0}\right\| \tag{7}$$

In the initial process of distinguishing positive and negative samples, this paper can assume that the probability for both positive and negative samples is 0.5, so $p = 0.5$. In this case, the gradient norm can be further simplified to Equation 8

$$\left\|\frac{\partial L}{\partial\theta^0}\right\| = 2s\left|N^f - N^b\right|\cdot\left\|\frac{\partial p}{\partial\theta^0}\right\| \tag{8}$$

The proof of gradient affecting distinguishing positive and negative samples for region proposal-based detectors, this paper extends the proof to keypoint-based detectors. For keypoint-based deep learning detectors, the loss function L is defined as Equation 9

$$\begin{aligned}L = \sum_{x=0}^{C}\Bigg[-s\sum_{i}^{N}\big[&y_{(i,j)}(x)\log(p_{(i,j)}(x))\\ &+(1 - y_{(i,j)}(x))\log(1 - p_{(i,j)}(x))\big]\Bigg]\end{aligned} \tag{9}$$

The gradient of the loss function for the parameters is calculated as Equation 10

$$\begin{aligned}\frac{\partial L}{\partial\theta^0} &= \sum_{x=0}^{C}\Bigg[-s\sum_{i}^{N}\Bigg[y_{(i,j)}(x)\frac{\partial\log(p_{(i,j)}(x))}{\partial\theta^0}\\ &\qquad +(1 - y_{(i,j)}(x))\frac{\partial\log(1 - p_{(i,j)}(x))}{\partial\theta^0}\Bigg]\Bigg]\\ &= \sum_{x=0}^{C}\Bigg[-s\sum_{i}^{N}\Bigg[y_{(i,j)}(x)\cdot\frac{1}{p_{(i,j)}(x)}\cdot\frac{\partial p_{(i,j)}(x)}{\partial\theta^0}\\ &\qquad -(1 - y_{(i,j)}(x))\cdot\frac{1}{1 - p_{(i,j)}(x)}\cdot\frac{\partial p_{(i,j)}(x)}{\partial\theta^0}\Bigg]\Bigg]\end{aligned} \tag{10}$$

The gradient norm is calculated as Equation 11

$$\left\|\frac{\partial L}{\partial\theta^0}\right\| = \sum_{x=0}^{C}\left[s\left|N^{x+}\frac{1}{p} - N^{x-}\frac{1}{1 - p}\right|\cdot\left\|\frac{\partial p}{\partial\theta^0}\right\|\right] \tag{11}$$

In the initial process of distinguishing positive and negative samples, the probability for both positive and negative samples is 0.5, so this paper assumes $p = 0.5$. In this case, the gradient norm can be further simplified to Equation 12

$$\left\|\frac{\partial L}{\partial\theta^0}\right\| = \sum_{x=0}^{C}\left[2s\left|N^{x+} - N^{x-}\right|\cdot\left\|\frac{\partial p}{\partial\theta^0}\right\|\right] \tag{12}$$

Finally, the sum of the positive samples $(x+)$ and the negative samples $(x-)$ equals the total number of samples as shown in Equation 13

$$(x+) + (x-) = sum\ of\ sampling \tag{13}$$

These equations delineate the loss function and its gradient computation within the gradient descent algorithm, specifically addressing the influence of gradient magnitude on the stability of model training. The magnitude of the gradient substantially affects the training process. A larger gradient results in a higher error rate due to the propensity for excessively large update steps, which compromise the stability and efficacy of the training process [43].

It can be concluded that the gradient value is related to the difference between positive and negative samples. For region proposal-based deep learning detectors, the gap between positive and negative samples is large. Usually, both region proposal-based and keypoint-based deep learning detectors divide the image into $k \times k$ grids, which is shown in Figure 2. The region proposal-based deep learning detectors generate b candidate bounding boxes in each grid, while the region proposal-based deep learning detectors generate a centerpoint or keypoint in each grid. For example, $k \times k$ grids generate $k \times k \times b$ bounding boxes, and the most suitable sample is only $\frac{1}{k\times k\times b}$. While for detectors based on key points or center points, the most suitable sample in $k \times k$ grids is $\frac{1}{k\times k}$, and the ratio of positive and negative samples is significantly reduced. However, most region proposal-based deep learning detectors can utilize the two-stage classification method to solve the class imbalance problem. They can classify background and foreground firstly and then classify categories of foregrounds. But the keypoint-based detectors do not have proposals, they can not use the two-stage method, so keypoint-based detectors suffer more from class imbalance.

This paper studies the optimization of keypoint-based detectors. Because keypoint-based detectors are usually one-stage detectors, the background and various foregrounds are classified together, resulting in sample imbalance. This paper has found that the greater the sum of a specific class, the easier it is to train because the gradient value is smaller. The number of foreground samples is much smaller than that of background samples, so background samples are the easiest to train. To reduce the impact of the background on the foreground, the weight of the background needs to be reduced during training.

There are two principal strategies for weight reduction of background, i.e., diminishing the weight of the over-trained samples and increasing the weight of the infrequent samples. In scenarios where background samples overwhelmingly dominate, employing frequency-based weighting initially results in an excessively low background weight. Given the abundance and ease of training background samples, leveraging weights based on training difficulty allows for expedited training and subsequent weight reduction of the background, thereby shifting the training focus to the foreground.

The foreground typically encompasses diverse classes with relatively scarce occurrences, resulting in a minor overall proportion. The training difficulty-based method is inadequate for handling scenarios with numerous classes and low proportions, because the rare-class objects will reduce the weights after training. After rare-class objects finish training, the deep learning model will ignore rare-class objects. When deep learning models focus on frequent-class objects, the performance on rare-class objects may become worse again. Consequently, utilizing frequency-based weights ensures that each class maintains fixed training weights, preventing the under-training of classes with insufficient samples.

B. ADAPTIVE GRADIENT FUNCTION

This paper has found that when $\left|N^{x+} - N^{x-}\right|$ approaches zero, the gradient becomes very small. However, the difference between N^{x+} and N^{x-} is unknown. So, this paper designs a new activation function to calculate the probability of positive and negative samples together, which is inspired by Class-Wise Difficulty-Balanced Loss [26]. Firstly, this paper defines $\bar{s_p}$, which can add the sum of positive samples when calculating probabilities of negative samples. $\bar{s_p}$ is the average input for all the p positive samples, which can be calculated as Equation 14

$$\bar{s_p} = \frac{\lambda}{N_{pos}} \sum_{i=1}^{N_{pos}} s_i \tag{14}$$

Here, N_{pos} is the number of the positive samples and s_i is the corresponding input of the activation function of the positive samples. And then this paper proposes the Adaptive Gradient Function to be the activation function of the last layer instead of the traditional Sigmoid function [7], which is as follows Equation 15

$$\begin{aligned} p_i &= \frac{e^{s_i}}{e^{s_i} + 1} \\ p_j &= \frac{e^{s_j}}{e^{s_j} + e^{\bar{s_p}}} \end{aligned} \tag{15}$$

Here, s_j is the corresponding input of the activation function of the negative samples. By introducing the Adaptive Gradient Function, this paper recalculates the gradients and gets the following Equation 16

$$\begin{aligned} \frac{\partial p_i}{\partial s_i} &= p_i(1 - p_i) \\ \frac{\partial p_j}{\partial s_j} &= p_j(1 - p_j) \\ \frac{\partial p_j}{\partial s_i} &= -\frac{\lambda}{N_{pos}} p_j(1 - p_j) \end{aligned} \tag{16}$$

To set a suitable value of λ, this paper assumes $\frac{\partial L}{\partial \theta^0} = 0$ at the beginning of training. The gradient $\frac{\partial L}{\partial \theta^0}$ is Equation 17

$$\begin{aligned} \frac{\partial L}{\partial \theta^0} &= -s \sum_{i}^{N} \left[y_i \frac{\partial \log(p_i)}{\partial \theta^0} + (1 - y_j) \frac{\partial \log(1 - p_j)}{\partial \theta^0} \right] \\ &= -s \sum_{i}^{N} \left[y_i \cdot \frac{1}{p_i} \cdot \frac{\partial p_i}{\partial \theta^0} - (1 - y_j) \cdot \frac{1}{1 - p_j} \cdot \frac{\partial p_j}{\partial \theta^0} \right] \\ &= -s \sum_{i}^{N} \left[y_i \cdot \frac{1}{p_i} \cdot \frac{\partial p_i}{\partial s_i} \cdot \frac{\partial s_i}{\partial \theta^0} \right. \\ &\quad \left. -(1 - y_j) \cdot \frac{1}{1 - p_j} \cdot \left(\frac{\partial p_j}{\partial s_i} \cdot \frac{\partial s_i}{\partial \theta^0} + \frac{\partial p_j}{\partial s_j} \cdot \frac{\partial s_j}{\partial \theta^0} \right) \right] \\ &= -2s \sum_{i}^{N} \left[\frac{\partial p_i}{\partial s_i} \cdot \frac{\partial s_i}{\partial \theta^0} - \left(\frac{\partial p_j}{\partial s_i} \cdot \frac{\partial s_i}{\partial \theta^0} + \frac{\partial p_j}{\partial s_j} \cdot \frac{\partial s_j}{\partial \theta^0} \right) \right] \end{aligned} \tag{17}$$

Initially, this paper assumes $p_i \approx p_j \approx 0.5$, and $y_i = 1, y_j = 0$. Where because of at the beginning, assume $\frac{\partial s_i}{\partial \theta^0} \approx \frac{\partial s_j}{\partial \theta^0} \approx \frac{\partial s}{\partial \theta^0}$, and then this paper can get Equation 18

$$\begin{aligned} \frac{\partial L}{\partial \theta^0} &= -2s \cdot \sum_{i}^{N} \left[\frac{\partial p_i}{\partial s_i} \cdot \frac{\partial s}{\partial \theta^0} - \left(\frac{\partial p_j}{\partial s_i} \cdot \frac{\partial s}{\partial \theta^0} + \frac{\partial p_j}{\partial s} \cdot \frac{\partial s}{\partial \theta^0} \right) \right] \\ &= -2s \cdot \sum_{i}^{N} \left[\frac{\partial p_i}{\partial s_i} - \frac{\partial p_j}{\partial s_j} - \frac{\partial p_j}{\partial s_i} \right] \cdot \frac{\partial s}{\partial \theta^0} \\ &= -2s \cdot \sum_{i}^{N} \left[p_i(1 - p_i) - p_j(1 - p_j) \right. \\ &\quad \left. + \frac{\lambda}{N_{pos}} p_j(1 - p_j) \right] \cdot \frac{\partial s}{\partial \theta^0} \\ &= -8s \cdot \left[N_{pos} - N_{neg} + \lambda \cdot \frac{N_{neg}}{N_{pos}} \right] \cdot \frac{\partial s}{\partial \theta^0} \end{aligned} \tag{18}$$

Therefore, to make the gradient small, $N_{pos} - N_{neg} + \lambda \cdot \frac{N_{neg}}{N_{pos}} = 0$ and then set $\lambda = \frac{N_{pos}}{N_{neg}} \cdot (N_{neg} - N_{pos})$.

Equation 18 presents the proof process for the activation function coefficient λ, demonstrating that this coefficient can effectively reduce the gradient of the cross-entropy loss function. Equations 19 and 20 merely add weighting factors to the cross-entropy function but fundamentally remain cross-entropy functions. Since the activation function is computed prior to the loss function, when deriving the coefficient λ in Equation 18, we assume no additional weighting factors. Therefore, the loss function in Equation 18 represents a simplified version of Equations 19 and 20.

This paper conducted a statistical analysis of the gradients for positive and negative samples during the training of RetinaNet on MS COCO, as shown in Figure 3, across four different sampling methods. CE exhibits significant gradient disparities between positive and negative samples throughout the training process. While FL and CBL mitigate these differences to some extent, noticeable imbalances still persist. However, our proposed adaptive gradient function continuously adjusts the gradients during training, maintaining a balanced gradient distribution between positive and negative samples, ensuring more stable and effective training.

C. THE PROPOSED ADAPTIVE FOCAL LOSS

First of all, this paper extends FL to solve class imbalance of keypoint-based deep learning detectors [4]. The new formula is Equation 19

$$\begin{aligned} FL(o, \hat{o}) = &-\sum_{i,j}^{k,k}[\alpha_{+1}(1-o)^{\gamma}1\{\hat{o}=1\}log(o) \\ &+\alpha_{-1}(o)^{\gamma}1\{\hat{o}=0\}log(1-o)], \\ &o = o(x,y), \hat{o} = \hat{o}(x,y), \\ &\alpha_{-1} + \alpha_{+1} = 1 \end{aligned} \tag{19}$$

Here, (i, j) represents the position on the output map. $p_{(i,j)}$ is a vector representing the classification prediction at position (i, j), and $y_{(i,j)}$ is a vector representing the classification truth value at the same position. $o(x, y)$ and $\hat{o}(x, y)$ represent the predicted value and true value of the probability for foreground, which is a scalar to classify background and foreground.

By reducing the loss contribution of easily-trained samples, FL enables the model to focus its learning resources on samples that are more difficult to correctly train. When facing with extremely imbalanced datasets, FL helps the model improve its ability to identify minority categories by dynamically adjusting the loss function. However, FL does not take into account the huge differences in the number of different-class samples. It will over-train individual samples that are high in frequency but difficult to train, thereby ignoring some rare samples.

Then, this paper extends CBL to solve class imbalance of keypoint-based deep learning detectors [27]. The new formula is Equation 20

$$\begin{aligned} CBL(p_{(i,j)}, y_{(i,j)}) &= \sum_{i,j}^{k,k} w_{(i,j)} \cdot CE(p_{(i,j)}, y_{(i,j)}) \\ &= -\sum_{i,j}^{k,k} w_{(i,j)} \cdot \sum_{c}^{C}[1\{y_{(i,j)}(c)=1\}log(p_{(i,j)}(c)) \\ &\quad + 1\{y_{(i,j)}(c)=0\}log(1-p_{(i,j)}(c))], \\ w_{(i,j)} &= \frac{1-\beta}{1-\beta^{n_{(i,j)}}} \end{aligned} \tag{20}$$

Here, (i, j) represents the position on the output map. $p_{(i,j)}$ is a vector representing the classification prediction at position (i, j), and $y_{(i,j)}$ is a vector representing the classification truth value at the same position.

In formula 20, n is the number of samples for each category, and β is a hyperparameter between 0 and 1 (usually close to 1), which controls the impact of sample size on the effective sample size. This formula balances the contribution of each category to the loss function by reducing the weight of high-frequency categories and increasing the weight of low-frequency categories.

CBL significantly improves the performance on multiple public datasets with significant class imbalance. It not only enhances the recognition rate of minority categories but also maintains or improves the overall performance of the model. However, it does not consider the difference between easy-to-train and hard-to-train, resulting in some high-frequency samples being ignored.

Deep learning detectors face significant challenges due to the imbalance between frequent and rare samples, as well as between easy-to-train and hard-to-train samples. In a two-stage detector, the process involves classifying the background and foreground first and then classifying the object categories. However, in the one-stage keypoint-based detector, the background and object categories are classified simultaneously, which exacerbates class imbalance issues. From experiments with FL and CBL, it is observed that foreground-background classification is influenced more by training difficulty, while object classification of different categories is influenced more by sample frequency.

As illustrated in Figure 4, the visualization shows three different sampling methods. Different colors represent various sample types, while different shapes indicate the varying difficulty levels of these samples. CE treats all samples equally, without distinguishing between easy or difficult ones. FL, on the other hand, allocates more resources to training difficult-to-train samples. CBL focuses on investing more resources in training rare-class samples. Our proposed AFL considers both scenarios simultaneously, leading to improved performance by effectively balancing the attention given to both difficult and rare-class samples.

To address these issues, this paper proposes a novel loss function called Adaptive Focal Loss (AFL). The name 'Adaptive Focal Loss' consists of two parts: 'Adaptive' is inspired by the Adaptive gradient function and

FIGURE 3. Sum of gradients for positive and negative samples during the training of RetinaNet on MS COCO dataset.

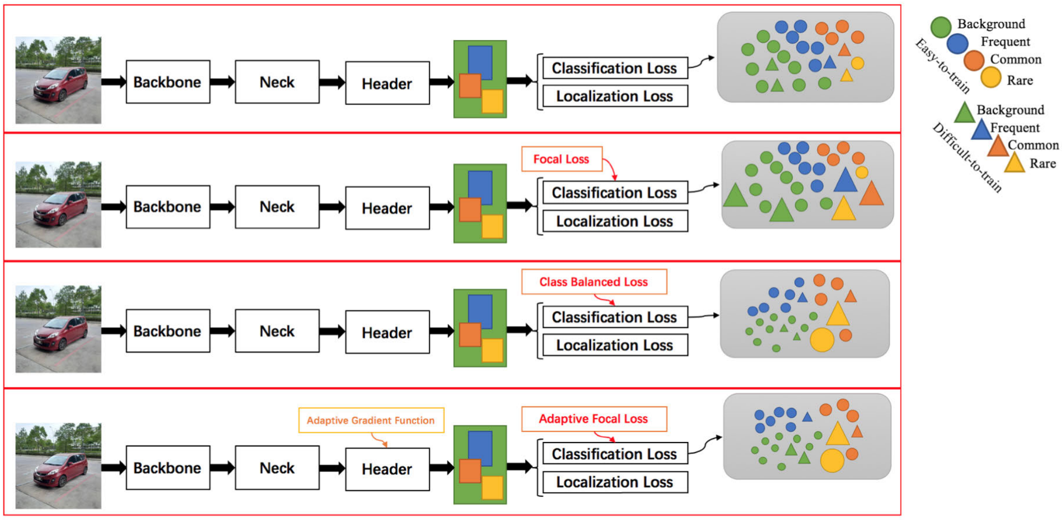

FIGURE 4. Visualization of different sampling methods.

Class-Balanced Loss [27], while 'Focal Loss' is derived from the original Focal Loss [4]. This method initially increases the loss weight for foreground-background discrimination to reduce the influence of the background class. It then shifts the focus to the under-trained rare classes, significantly enhancing detection performance.

$$L_{AFL} = L_{b-f} + L_{obj} \tag{21}$$

After training to distinguish foreground from background, it is necessary to reduce the impact of over-trained objects, as shown in Equation 22.

$$L_{b-f} = FL(o, \hat{o}) \tag{22}$$

In Equation 19, $FL(o, \hat{o})$ represents the focal loss, where o indicates whether a sample belongs to the foreground. In Equation 22, L_{b-f} stands for the loss associated with predicting whether a sample is background or foreground. Since the foreground inherently represents objects, the paper uses o to denote it. Additionally, during the training process, both ground truth and predicted values are required, which is why the paper uses $\hat{o}$ to represent the ground truth and o to represent the predicted value.

To increase the training weight of the rare-class objects, the loss function at each position needs to be re-weighted as Equation 23

$$L_{obj} = CBL(p_{(i,j)}, y_{(i,j)}) \tag{23}$$

The loss function for object detection comprises two main components: $L_{\text{bounding box}}$, which handles bounding box regression, and $L_{\text{classification}}$, which addresses the classification task. Therefore, the classification loss L_{cls} is given by

Equation 24

$$
\begin{aligned}
L_{cls} &= L_{f-b} + L_{obj} \\
&= -\sum_{i,j}^{k,k}[\alpha_{+1}(1-o(x,y))^{\gamma}1\{\hat{o}(x,y)=1\}log(o(x,y)) \\
&\quad + \alpha_{-1}(o(x,y))^{\gamma}1\{\hat{o}(x,y)=0\}log(1-o(x,y))] \\
&\quad - \sum_{i,j}^{k,k} w_{(i,j)} \cdot \sum_{c}^{C}[1\{y_{(i,j)}(c)=1\}log(p_{(i,j)}(c)) \\
&\quad + 1\{y_{(i,j)}(c)=0\}log(1-p_{(i,j)}(c))], \\
w_{(i,j)} &= \frac{1-\beta}{1-\beta^{n(i,j)}}, \\
&\quad \alpha_{-1}+\alpha_{+1}=1 \qquad (24)
\end{aligned}
$$

As shown in Equation 24, (i, j) represents the position on the output map. $p_{(i,j)}$ is a vector representing the classification prediction at position (i, j), and $y_{(i,j)}$ is a vector representing the truth classification at the same position. The length of both $p_{(i,j)}$ and $y_{(i,j)}$ is C. This paper assumes $y_{(i,j)}(c) \in \{y_{(i,j)}(1), y_{(i,j)}(2), y_{(i,j)}(3), \ldots, y_{(i,j)}(C)\}$, which represents the true probability of the specific class. This paper assumes $p_{(i,j)}(c) \in \{p_{(i,j)}(1), p_{(i,j)}(2), p_{(i,j)}(3), \ldots, p_{(i,j)}(C)\}$, which represents the predicted probability of the specific class.

IV. EXPERIMENTS

The experimental results presented in the following tables compare different sampling methods for state-of-the-art keypoint-based detectors, including CenterNet, FCOS, RetinaNet, ExtremeNet, CornerNet, and ObjectBox, across three widely used datasets: LVIS, MS COCO, and PASCAL VOC. The visualization of the experimental results is shown in Figure 5. Besides, an ablation study was conducted to determine the optimal values for the three hyperparameters of AFL, specifically β, γ, and α_{+1}. The visualization of the ablation study is shown in Figure 9. Additionally, All input images for these detectors were resized to 600 × 600 and evaluated on Tianan GPUs.

A. EXPERIMENTAL RESULTS OF DIFFERENT SAMPLING METHODS ON LVIS DATASET

The experimental results demonstrate a significant variation in the performance of different sampling methods within the FCOS framework on the LVIS dataset. AFL achieves the highest overall performance, with an AP of 27.9, $AP_{0.5}$ of 42.8, and superior results across rare, common, and frequent categories. This indicates that AFL effectively balances the contributions of all classes, outperforming traditional sampling strategies.

Methods such as BAGS, BALMS, and EQLv2 show relatively strong performance, with AP values ranging between 27.0 and 27.2. These approaches leverage various balancing mechanisms to improve detection performance, particularly for underrepresented categories. However, their results are consistently lower than those of AFL, highlighting the advantages of AFL's adaptive hyperparameter tuning.

Conventional sampling methods, including CE, LWS, and LDAM, achieve lower AP values, approximately 26.0–26.2. While these methods are simpler and computationally efficient, they struggle to address the inherent class imbalance in the LVIS dataset, leading to suboptimal detection performance for rare and common categories.

Interestingly, FL achieves slightly better performance than most conventional methods, with an AP of 26.5. However, it still falls short compared to advanced methods like AFL, emphasizing the importance of tailored loss functions and dynamic hyperparameter adjustments in handling imbalanced datasets. The FPS remains consistent across all methods, demonstrating the computational efficiency of the sampling techniques.

TABLE 1. Comparisons of different sampling methods in FCOS framework on LVIS dataset ($AP = AP_{0.5\sim0.95}$).

Loss	AP	$AP_{0.5}$	AP_r	AP_c	AP_f	FPS
CE [41]	26.0	40.6	18.7	29.9	25.6	∼4
LWS [42]	26.2	40.6	18.7	30.0	25.9	∼4
LDAM [43]	26.0	40.7	18.7	30.1	25.6	∼4
EQL [21]	26.1	40.8	18.7	30.0	25.6	∼4
Forest R-CNN [44]	26.2	40.8	18.8	29.9	25.7	∼4
RFS [45]	26.1	41.0	18.7	29.9	25.7	∼4
BAGS [46]	27.0	41.6	19.7	30.9	26.6	∼4
BALMS [47]	27.1	41.6	19.8	30.9	26.7	∼4
EQLv2 [48]	27.1	41.6	19.9	31.0	26.8	∼4
DisAlign [49]	27.2	41.6	19.7	30.9	26.6	∼4
LOCE [9]	27.0	41.7	19.7	31.1	26.7	∼4
CBL [27]	26.2	40.7	18.7	30.0	25.7	∼4
FL [4]	26.5	41.4	19.3	28.6	26.6	∼4
AFL	**27.9**	**42.8**	**20.7**	**29.9**	**27.9**	∼4

B. EXPERIMENTAL RESULTS ON LVIS DATASET

The experimental results on the LVIS dataset highlight the consistent performance improvements achieved by using the AFL loss across various detectors. CenterNet, for example, shows a significant boost in AP, increasing from 25.1 with CE loss to 28.2 with AFL, particularly excelling in rare categories (AP_r), which reflects AFL's ability to address class imbalance effectively.

For FCOS, the trend is similar, where AFL outperforms other loss functions, achieving the highest AP of 27.9 compared to 24.9 with CE and 26.4 with FL. The improvements in both AP_r and AP_m further confirm AFL's efficacy in enhancing detection performance for challenging scenarios.

RetinaNet also benefits from AFL, achieving its highest AP of 14.5. Despite starting from a relatively lower baseline compared to other detectors, the improvements in AP_c and AP_f underline AFL's robustness and adaptability across varying feature complexities.

ExtremeNet and CornerNet exhibit similar trends, with AFL consistently delivering the best results in AP and its subcategories. For ExtremeNet, the AP_r and AP_m metrics see noticeable gains, while CornerNet achieves its best performance with AFL, reaching an AP of 23.8 compared to 21.8 with CE.

Finally, ObjectBox demonstrates AFL's superiority, with the highest AP of 29.6, particularly excelling in medium and large object detection (AP_m and AP_l). These results showcase AFL's potential in handling diverse object scales and complex detection tasks.

TABLE 2. Comparisons of different sampling methods on LVIS dataset ($AP = AP_{0.5\sim0.95}$).

Detector	Loss	AP	$AP_{0.5}$	AP_r	AP_c	AP_f	FPS
CenterNet [2]	CE [41]	25.1	39.4	15.0	27.0	27.2	~3
CenterNet [2]	CBL [27]	26.5	41.0	17.9	29.0	27.3	~3
CenterNet [2]	FL [4]	24.7	38.7	13.4	26.5	27.3	~3
CenterNet [2]	AFL	**28.2**	**45.0**	**21.0**	**31.3**	**27.2**	~3
FCOS [5]	CE [41]	24.9	38.8	13.5	26.8	27.5	~ 4
FCOS [5]	CBL [27]	25.2	39.3	14.4	27.4	27.6	~ 4
FCOS [5]	FL [4]	26.4	40.6	17.1	28.8	27.3	~ 4
FCOS [5]	AFL	**27.9**	**42.8**	**20.7**	**29.9**	**27.9**	~ 4
RetinaNet [4]	CE [41]	11.3	23.3	2.2	14.4	21.7	12
RetinaNet [4]	CBL [27]	11.6	23.8	2.9	15.0	21.8	12
RetinaNet [4]	FL [4]	13.1	25.2	3.5	18.4	19.2	12
RetinaNet [4]	AFL	**14.5**	**27.1**	**3.9**	**19.3**	**19.7**	12
ExtremeNet [3]	CE [41]	20.4	30.2	6.3	22.5	25.0	~4
ExtremeNet [3]	CBL [27]	21.8	31.9	8.3	24.3	26.1	~4
ExtremeNet [3]	FL [4]	22.1	32.6	10.7	27.4	25.1	~4
ExtremeNet [3]	AFL	**23.7**	**29.1**	**13.5**	**27.9**	**25.2**	~4
CornerNet [1]	CE [41]	21.8	31.1	5.0	25.4	26.2	~3
CornerNet [1]	CBL [27]	22.6	32.4	6.8	26.0	27.0	~3
CornerNet [1]	FL [4]	23.0	33.1	8.0	26.4	27.4	~3
CornerNet [1]	AFL	**23.8**	**35.6**	**10.0**	**27.1**	**26.1**	~3
ObjectBox [6]	CE [41]	27.0	40.6	12.7	28.8	29.5	~6
ObjectBox [6]	CBL [27]	27.7	41.6	14.0	30.0	30.0	~6
ObjectBox [6]	FL [4]	29.1	42.0	16.8	33.1	29.0	~6
ObjectBox [6]	AFL	**29.6**	**44.2**	**19.2**	**33.3**	**29.2**	~6

C. EXPERIMENTAL RESULTS ON MS COCO DATASET

The results presented in Table 3 demonstrate that the proposed AFL consistently outperforms other sampling methods, including Original Cross Entropy (CE), CBL, and FL, across multiple detectors on the MS COCO dataset. Specifically, AFL achieves the highest AP, $AP_{0.5}$, AP_s, AP_m, and AP_l metrics for each detector, indicating its superior ability to handle class imbalance and improve detection accuracy. Besides, detection results of FL and AFL in FCOS on LVIS dataset are shown in Figure 7.

For example, in the CenterNet detector, AFL improves AP to 46.8%, which is higher than that achieved with FL (45.4%) and significantly better than CE (44.9%). Similar trends are observed with FCOS, where AFL increases AP to 46.6%, outperforming FL (45.2%) and CE (44.7%).

In RetinaNet, the performance gains are particularly notable, with AFL raising AP to 33.2%, compared to 31.9% with FL and 31.1% with CE. ExtremeNet and CornerNet detectors also benefit from AFL, showing improvements across all evaluated metrics.

Finally, in the ObjectBox detector, AFL achieves an AP of 48.3%, surpassing the results of FL (47.9%) and CE (46.8%). As shown in Figure 5, these consistent improvements across various detectors and evaluation metrics underscore the effectiveness of AFL in enhancing object detection performance in challenging scenarios.

TABLE 3. Comparisons of different sampling methods on MS COCO dataset ($AP = AP_{0.5\sim0.95}$).

Detector	Loss	AP	$AP_{0.5}$	AP_s	AP_m	AP_l	FPS
CenterNet [2]	CE [41]	44.9	62.4	25.6	47.4	57.4	~3
CenterNet [2]	CBL [27]	45.1	62.5	25.8	47.5	57.6	~3
CenterNet [2]	FL [4]	45.4	62.9	25.8	48.1	57.9	~3
CenterNet [2]	AFL	**46.8**	**64.1**	**27.2**	**49.2**	**59.3**	~3
FCOS [5]	CE [41]	44.7	64.1	27.6	47.5	55.6	~4
FCOS [5]	CBL [27]	44.9	64.2	27.6	47.6	55.7	~4
FCOS [5]	FL [4]	45.2	64.9	28.2	46.2	56.6	~4
FCOS [5]	AFL	**46.6**	**66.3**	**29.6**	**47.5**	**57.9**	~4
RetinaNet [4]	CE [41]	31.1	48.6	10.9	35.1	49.8	12
RetinaNet [4]	CBL [27]	31.3	48.7	10.9	35.2	49.9	12
RetinaNet [4]	FL [4]	31.9	49.5	11.6	35.8	48.5	12
RetinaNet [4]	AFL	**33.2**	**50.6**	**12.8**	**36.9**	**49.7**	12
ExtremeNet [3]	CE [41]	40.2	55.5	20.4	43.2	53.1	~4
ExtremeNet [3]	CBL [27]	41.5	56.8	21.5	44.5	54.0	~4
ExtremeNet [3]	FL [4]	41.7	56.9	21.8	44.8	54.2	~4
ExtremeNet [3]	AFL	**42.4**	**52.6**	**22.4**	**45.5**	**55.2**	~4
CornerNet [1]	CE [41]	40.6	56.4	19.1	42.8	54.3	~3
CornerNet [1]	CBL [27]	41.4	57.3	20.0	43.6	55.1	~3
CornerNet [1]	FL [4]	41.7	57.4	20.2	43.8	55.5	~3
CornerNet [1]	AFL	**42.5**	**58.2**	**21.1**	**44.7**	**56.1**	~3
ObjectBox [6]	CE [41]	46.8	65.9	26.8	49.5	57.6	~6
ObjectBox [6]	CBL [27]	47.4	66.5	27.2	50.1	58.0	~6
ObjectBox [6]	FL [4]	47.9	66.3	27.9	50.5	58.3	~6
ObjectBox [6]	AFL	**48.3**	**67.7**	**28.1**	**50.9**	**59.2**	~6

D. EXPERIMENTAL RESULTS ON PASCAL VOC DATASET

The results in Table 5 demonstrate the performance of different sampling methods on the PASCAL VOC 2007 dataset. Across all detectors, the AFL method consistently achieves the highest $AP_{0.5}$ scores, indicating its effectiveness in improving detection performance. For instance, CenterNet achieves an $AP_{0.5}$ of 89.8 with AFL, compared to 89.6 with CE and CBL, and 89.7 with FL, highlighting AFL's slight but consistent improvement.

FCOS shows similar trends, with AFL improving its $AP_{0.5}$ to 88.6, surpassing CE and CBL at 87.3 and FL at 87.5. This improvement suggests that AFL enhances FCOS's ability to detect objects more accurately, especially in challenging scenarios where small performance gains can be critical.

RetinaNet exhibits a noticeable performance gain with AFL, reaching an $AP_{0.5}$ of 77.2, compared to 76.8 with CE and FL and 76.9 with CBL. This trend underscores AFL's adaptability and potential to boost the performance of detectors with different architectures.

ExtremeNet and CornerNet also benefit from AFL, achieving $AP_{0.5}$ scores of 82.6 and 82.7, respectively, slightly outperforming other methods. For ObjectBox, AFL achieves the highest $AP_{0.5}$ score of 91.8, further demonstrating its capability to enhance object detection across various detectors. Despite these improvements, the FPS remains consistent across sampling methods for each detector, ensuring no compromise in efficiency.

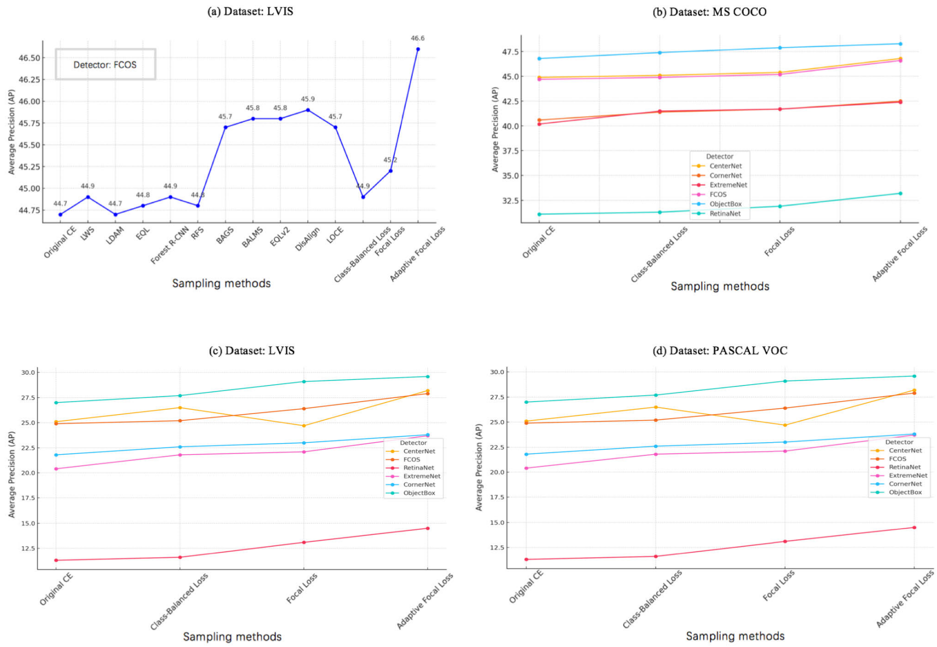

FIGURE 5. **Visual Comparisons of different sampling methods on different dataset.**

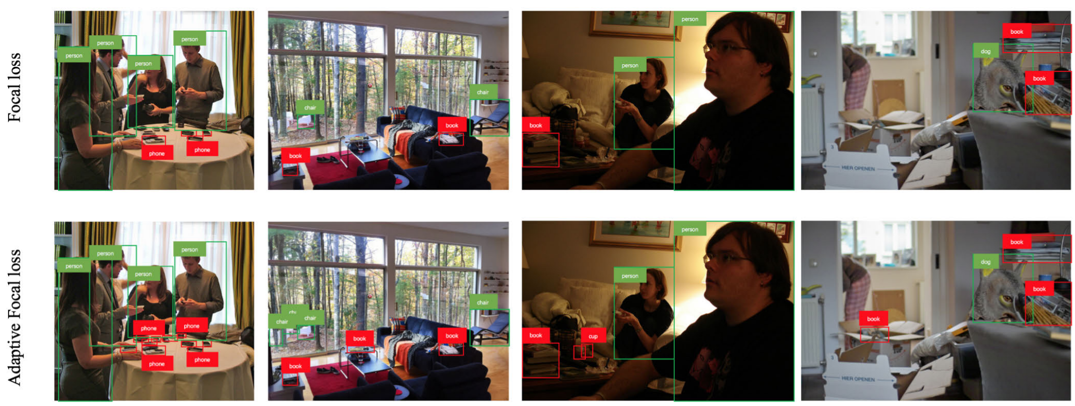

FIGURE 6. **Detection results of FL and AFL in FCOS on LVIS dataset.**

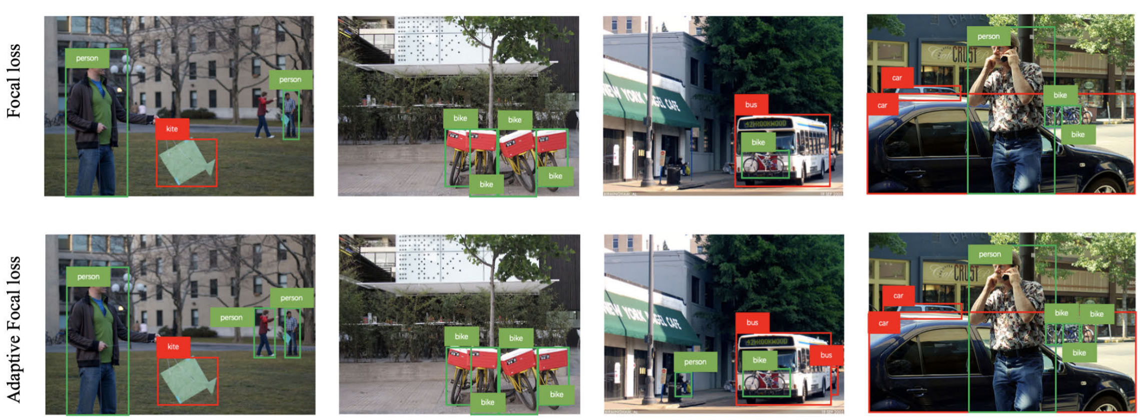

FIGURE 7. **Detection results of FL and AFL in FCOS on MS COCO dataset.**

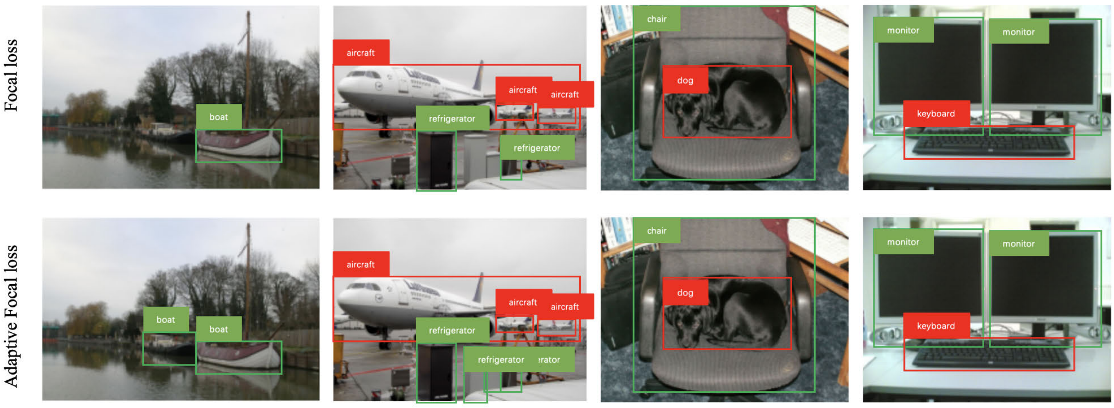

FIGURE 8. Detection results of FL and AFL in FCOS on PASCAL VOC dataset.

TABLE 4. Comparisons of different sampling methods on PASCAL VOC dataset.

Detector	Loss	$AP_{0.5}$	FPS
CenterNet [2]	CE [41]	89.6	~3
CenterNet [2]	CBL [27]	89.6	~3
CenterNet [2]	FL [4]	89.7	~3
CenterNet [2]	AFL	**89.8**	~3
FCOS [5]	CE [41]	87.3	~4
FCOS [5]	CBL [27]	87.3	~4
FCOS [5]	FL [4]	87.5	~4
FCOS [5]	AFL	**88.6**	~4
RetinaNet [4]	CE [41]	76.8	12
RetinaNet [4]	CBL [27]	76.9	12
RetinaNet [4]	FL [4]	76.8	12
RetinaNet [4]	AFL	**77.2**	12
ExtremeNet [3]	CE [41]	82.2	~4
ExtremeNet [3]	CBL [27]	82.3	~4
ExtremeNet [3]	FL [4]	82.4	~4
ExtremeNet [3]	AFL	**82.6**	~4
CornerNet [1]	CE [41]	82.5	~3
CornerNet [1]	CBL [27]	82.6	~3
CornerNet [1]	FL [4]	82.6	~3
CornerNet [1]	AFL	**82.7**	~3
ObjectBox [6]	CE [41]	91.5	~6
ObjectBox [6]	CBL [27]	91.5	~6
ObjectBox [6]	FL [4]	91.7	~6
ObjectBox [6]	AFL	**91.8**	~6

E. ABLATION STUDY

β, γ, and α_{+1} are the three hyperparameters of AFL. The parameter β ranges from 0 to 1 and dictates that as the frequency of a class increases, the training weight decreases. However, this difference in weight must be reasonable; otherwise, it can negatively affect both frequent and rare samples. If β exceeds 1, the effect is reversed, giving higher weight to frequent classes, which can exacerbate class imbalance. The parameter γ controls the training weight between difficult-to-train and easy-to-train classes. A larger γ causes the deep learning detector to focus more on difficult-to-train classes, but if γ is too high, the easy-to-train classes may be neglected, leading to poor results. The parameter α_{+1} controls the proportion of positive samples, with a range between 0 and 1. If α_{+1} is too low, deep learning detectors may completely ignore positive samples, while if it is too high, the detectors might entirely overlook negative samples. An optimal balance that slightly favors negative samples tends to yield the best outcomes.

1) ABLATION STUDY OF β

The ablation study results presented in Table 5 illustrate the impact of different values of β on the performance metrics of the MS COCO dataset after adjustments. The results show that as β increases from 0 to 0.5, there is a consistent improvement in all evaluated metrics, with the highest AP (46.6%), $AP_{0.5}$ (66.3%), AP_s (29.6%), AP_m (47.5%), and AP_l (57.7%) achieved at $\beta = 0.5$. This suggests that a moderate value of β optimally balances the model's ability to detect objects across different scales and complexity. However, as β continues to increase beyond 0.5, the performance metrics begin to decline, with a significant drop observed at $\beta = 5$ and $\beta = 10$, where AP decreases to 40.5% and 31.1%, respectively. These results indicate that while adjusting β can enhance model performance, excessively high values can lead to overfitting or reduced generalization, as reflected in the lower AP scores across all categories, which is shown in Figure 9. The FPS remains consistent at about 4 across all values of β, indicating that the changes in β primarily affect detection accuracy rather than inference speed.

TABLE 5. Ablation study of β on MS COCO dataset after adjustments ($AP = AP_{0.5\sim0.95}$).

β	AP	$AP_{0.5}$	AP_s	AP_m	AP_l	FPS
0	43.5	63.3	26.7	44.4	54.8	~4
0.1	45.1	64.6	28.1	46.2	56.5	~4
0.3	46.5	65.9	29.5	47.4	57.7	~4
0.5	**46.6**	**66.3**	**29.6**	**47.5**	**57.9**	~4
0.7	46.5	66.0	29.5	47.5	57.7	~4
0.9	46.4	66.0	29.4	47.3	57.6	~4
5	40.5	60.9	25.6	44.3	54.1	~4
10	31.1	51.9	19.6	36.3	43.8	~4

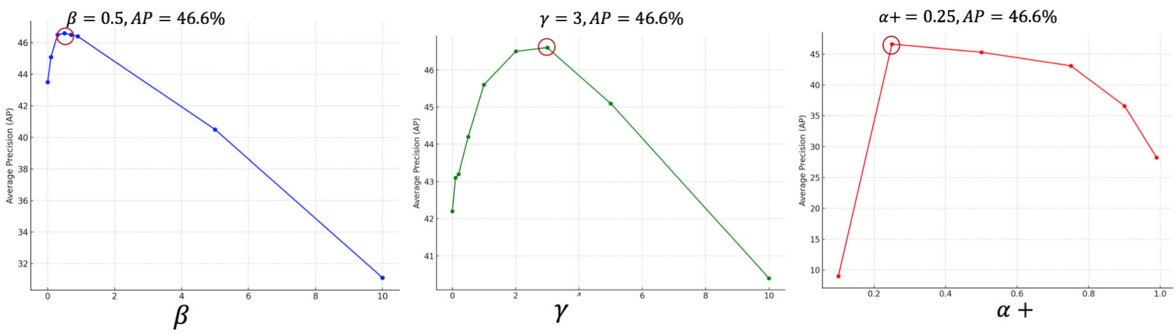

FIGURE 9. Visualization of ablation study on MS COCO dataset.

2) ABLATION STUDY OF γ

The ablation study results presented in Table 7 examine the effect of varying γ on the performance of a model trained on the MS COCO dataset. As shown, increasing γ from 0 to 3.0 leads to a steady improvement in all evaluation metrics, with the best performance achieved at $\gamma = 3.0$. Specifically, the AP rises to 46.6%, $AP_{0.5}$ to 66.3%, AP_s to 29.6%, AP_m to 47.5%, and AP_l to 57.9%. These improvements suggest that moderate values of γ effectively enhance the model's ability to detect objects across varying scales. When γ reaches 5.0 and 10.0, there is a notable decline in performance, with AP dropping to 45.1% and 40.4%, respectively, highlighting the risk of over-penalizing easy examples, which can degrade overall model accuracy, which is shown in Figure 9. The FPS remains consistent at about 4 across all settings, indicating that the variation in γ primarily impacts detection accuracy rather than computational efficiency.

TABLE 6. Ablation study of γ on MS COCO dataset after adjustments ($AP = AP_{0.5\sim0.95}$).

γ	AP	$AP_{0.5}$	AP_s	AP_m	AP_l	FPS
0	42.2	62.3	25.5	43.1	53.7	~4
0.1	43.1	62.7	26.1	44.2	54.5	~4
0.2	43.2	62.8	26.1	44.2	54.6	~4
0.5	44.2	63.9	27.1	45.3	55.4	~4
1.0	45.6	65.3	28.6	46.7	56.8	~4
2.0	46.5	66.1	29.5	47.5	57.7	~4
3.0	**46.6**	**66.3**	**29.6**	**47.5**	**57.9**	~4
5.0	45.1	64.9	28.2	46.3	56.2	~4
10.0	40.4	60.4	22.5	42.1	51.6	~4

3) ABLATION STUDY OF α

The ablation study results in Table 7 illustrate the impact of different values of α_{+1} on the performance of a model trained on the MS COCO dataset after adjustments. The results demonstrate that at $\alpha_{+1} = 0.1$, the model performs poorly, with negative AP values across all metrics, indicating a significant degradation in detection performance. As α_{+1} increases to 0.25, the model achieves its best performance, with AP reaching 46.6%, $AP_{0.5}$ at 66.3%, AP_s at 29.6%, AP_m at 47.5%, and AP_l at 57.9%, suggesting that this setting optimally balances the model's ability to detect objects of various sizes. However, further increases in α_{+1} lead to a gradual decline in performance. For example, at $\alpha_{+1} = 0.5$, the AP drops slightly to 45.3%, and this downward trend continues, with the most significant decrease observed at $\alpha_{+1} = 0.99$, where AP falls to 28.2%. These results indicate that while a moderate value of α_{+1} can enhance performance, excessively high values negatively impact the model's ability to generalize, which is shown in Figure 9. The FPS remains consistent at about 4 across all values, suggesting that the variations in α_{+1} primarily influence detection accuracy rather than computational efficiency.

TABLE 7. Ablation study of γ on MS COCO dataset after adjustments ($AP = AP_{0.5\sim0.95}$).

α+1	AP	$AP_{0.5}$	AP_s	AP_m	AP_l	FPS
0.1	9.0	7.3	10.1	11.0	10.9	~4
0.25	**46.6**	**66.3**	**29.6**	**47.5**	**57.9**	~4
0.5	45.3	64.9	28.3	46.4	56.8	~4
0.75	43.1	62.9	25.1	45.2	54.4	~4
0.9	36.6	55.4	19.4	39.1	47.6	~4
0.99	28.2	47.3	14.1	31.0	40.6	~4

F. ANALYSIS

The experimental results across the three datasets, i.e., MS COCO 2017, LVIS v5.0, and PASCAL VOC 2007, consistently demonstrate the superior performance of AFL compared to CE, CBL, and FL, as illustrated in Figure 5. CE applies equal treatment to all samples, which leads to significant gradient imbalances during training, particularly in scenarios with severe class imbalance, as reflected in its lower performance across all three datasets. CBL improves on this by adjusting loss weights based on class frequencies, giving more emphasis to rare classes, but it still struggles to effectively balance the learning process between easy and difficult samples. FL addresses this issue by prioritizing hard-to-train samples, enhancing performance in challenging detection tasks. However, it may still fail to sufficiently accommodate rare-class objects. In contrast, AFL integrates the strengths of both CBL and FL while introducing an adaptive gradient activation function that dynamically adjusts gradients during training. This design results in consistently higher *AP* scores across all datasets, with significant improvements in rare and challenging object detection, particularly on long-tailed datasets such as LVIS. The superior performance of AFL, as demonstrated in these experiments, confirms its effectiveness in addressing the inherent limitations of other sampling methods, providing a more balanced and efficient solution for managing class imbalance in deep learning detectors.

Nevertheless, AFL has certain limitations in addressing foreground-foreground and foreground-background imbalances. While AFL demonstrates substantial advantages over Focal Loss when applied to long-tailed datasets such as LVIS, its performance shows only marginal improvements on standard foreground-background imbalance datasets like PASCAL VOC, where the gains compared to Focal Loss are less pronounced. This limitation stems from AFL's strong dependence on the number of object categories, which can lead to redundant training of already learned rare classes, ultimately wasting computational resources during training.

V. CONCLUSION

In conclusion, this paper presents AFL, a novel approach designed to address the challenges of class imbalance in keypoint-based deep learning detectors. By analyzing the detrimental effects of large gradients during the early stages of training, this work introduces an Adaptive Gradient Function to mitigate these impacts, facilitating more effective learning. Building on the strengths of FL and CBL, AFL effectively balances the learning process across varying sample difficulties and class distributions. Extensive experiments demonstrate significant performance improvements across multiple datasets, including MS COCO, PASCAL VOC, and LVIS. Notably, AFL excels in rare-class object detection on the long-tailed LVIS dataset, underscoring its effectiveness in real-world applications.

Despite the promising results, some limitations should be acknowledged. The computational complexity of AFL, stemming from the integration of FL and CBL principles, can increase training time, particularly for large-scale datasets. Additionally, while AFL achieves remarkable performance in rare-class detection, its effectiveness on extremely small objects or highly occluded objects remains an area for further investigation. Future research should also explore the method's potential to generalize beyond object detection tasks, such as instance segmentation and human pose estimation.

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REFERENCES

[1] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jan. 2018, pp. 765–781.

[2] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6568–6577.

[3] X. Zhou, J. Zhuo, and P. Krähenbühl, "Bottom-up object detection by grouping extreme and center points," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 850–859.

[4] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[5] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: A simple and strong anchor-free object detector," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 1922–1933, Apr. 2022.

[6] M. Zand, A. Etemad, and M. Greenspan, "ObjectBox: From centers to boxes for anchor-free object detection," in *Proc. Eur. Conf. Comput. Vis.*, Jan. 2022, pp. 390–406.

[7] S. R. Dubey, S. K. Singh, and B. B. Chaudhuri, "Activation functions in deep learning: A comprehensive survey and benchmark," *Neurocomputing*, vol. 503, pp. 92–108, Sep. 2022.

[8] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5987–5995.

[9] C. Feng, Y. Zhong, and W. Huang, "Exploring classification equilibrium in long-tailed object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3397–3406.

[10] X. Xie, C. Lang, S. Miao, G. Cheng, K. Li, and J. Han, "Mutual-assistance learning for object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 15171–15184, Dec. 2023.

[11] Y. Cao, K. Chen, C. C. Loy, and D. Lin, "Prime sample attention in object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. pattern Recognit.*, pp. 11583–11591, 2020.

[12] C. Lang, G. Cheng, B. Tu, and J. Han, "Learning what not to segment: A new perspective on few-shot segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8047–8057.

[13] J. Chen, D. Liu, T. Xu, S. Wu, Y. Cheng, and E. Chen, "Is heuristic sampling necessary in training deep object detectors?" *IEEE Trans. Image Process.*, vol. 30, pp. 8454–8467, 2021.

[14] G. Cheng, C. Lang, and J. Han, "Holistic prototype activation for few-shot segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4650–4666, Apr. 2023.

[15] J. Chen, Q. Wu, D. Liu, and T. Xu, "Foreground-background imbalance problem in deep object detectors: A review," in *Proc. IEEE Conf. Multimedia Inf. Process. Retr. (MIPR)*, Aug. 2020, pp. 285–290.

[16] C. Lang, G. Cheng, B. Tu, C. Li, and J. Han, "Base and meta: A new perspective on few-shot segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10669–10686, Sep. 2023.

[17] C.-L. Duan, Y. Li, X.-S. Wei, and L. Zhao, "Long-tailed object detection pre-training: Dynamic rebalancing contrastive learning with dual reconstruction," 2024, *arXiv:2411.09453*.

[18] Y. Zhang, C. Zhang, L. Zhang, T. Liu, J. Guan, X. Liang, J. Zhao, and S. Zhou, "Tail classes matter: Long-tailed object detection revisited," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 3140–3144.

[19] J. Chen, B. Luo, Q. Wu, J. Chen, and X. Peng, "Overlap sampler for region-based object detection," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2020, pp. 756–764.

[20] S. Zhang, L. Wen, X. Bian, Z. Lei, and S. Z. Li, "Single-shot refinement neural network for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4203–4212.

[21] J. Tan, C. Wang, B. Li, Q. Li, W. Ouyang, C. Yin, and J. Yan, "Equalization loss for long-tailed object recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11659–11668.

[22] X. Li, W. Wang, L. Wu, S. Chen, X. Hu, J. Li, J. Tang, and J. Yang, "Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2020, pp. 21002–21012.

[23] J. Chen, D. Liu, B. Luo, X. Peng, T. Xu, and E. Chen, "Residual objectness for imbalance reduction," *Pattern Recognit.*, vol. 130, Oct. 2022, Art. no. 108781.

[24] T. Hao, Y. Tao, M. Li, X. Ma, P. Dong, L. Cui, P. Lv, and M. Xu, "Foreground and background separate adaptive equilibrium gradients loss for long-tail object detection," in *Proc. Int. Conf. Comput. Vis. Media*. Springer, Jan. 2024, pp. 200–218.

[25] Y. Zhao, S. Chen, S. Liu, Z. Hu, and J. Xia, "Hierarchical equalization loss for long-tailed instance segmentation," *IEEE Trans. Multimedia*, vol. 26, pp. 6943–6955, 2024.

[26] S. Sinha, H. Ohashi, and K. Nakamura, "Class-wise difficulty-balanced loss for solving class-imbalance," in *Proc. Asian Conf. Comput. Vis.*, Jan. 2021, pp. 549–565.

[27] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, "Class-balanced loss based on effective number of samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9260–9269.

[28] M. Shao and Z. Peng, "Distance metric-based learning for long-tail object detection," *Image Vis. Comput.*, vol. 142, Feb. 2024, Art. no. 104888.

[29] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common objects in context," in *Proc. 13th Eur. Conf. Comput. Vision (ECCV)*, Zurich, Switzerland, Sep. 2014, pp. 740–755.

[30] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, Jun. 2010.

[31] M. Everingham, S. M. A. Eslami, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes challenge: A retrospective," *Int. J. Comput. Vis.*, vol. 111, no. 1, pp. 98–136, Jan. 2015.

[32] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, Dec. 2015, pp. 91–99.

[33] K. Oksuz, B. C. Cam, E. Akbas, and S. Kalkan, "Rank & sort loss for object detection and instance segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 2989–2998.

[34] C. Zhang, G. Almpanidis, G. Fan, B. Deng, Y. Zhang, J. Liu, A. Kamel, P. Soda, and J. Gama, "A systematic review on long-tailed learning," 2024, *arXiv:2408.00483*.

[35] C. Du, Y. Wang, S. Song, and G. Huang, "Probabilistic contrastive learning for long-tailed visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 9, pp. 5890–5904, Sep. 2024.

[36] W. Liu, Z. Wu, Y. Wang, H. Ding, F. Liu, J. Lin, and G. Lin, "LCReg: Long-tailed image classification with latent categories based recognition," *Pattern Recognit.*, vol. 145, Jan. 2024, Art. no. 109971.

[37] L. Zhao, Y. Teng, and L. Wang, "Logit normalization for long-tail object detection," *Int. J. Comput. Vis.*, vol. 132, no. 6, pp. 2114–2134, Jun. 2024.

[38] J. Nie, R. M. Anwer, H. Cholakkal, F. S. Khan, Y. Pang, and L. Shao, "Enriched feature guided refinement network for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9536–9545.

[39] B. Li, Y. Liu, and X. Wang, "Gradient harmonized single-stage detector," in *Proc. AAAI Conf. Artif. Intell.*, Jan. 2018, pp. 8577–8584.

[40] Q. Cai, Y. Pan, Y. Wang, J. Liu, T. Yao, and T. Mei, "Learning a unified sample weighting network for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 14161–14170.

[41] Y. Ho and S. Wookey, "The real-world-weight cross-entropy loss function: Modeling the costs of mislabeling," *IEEE Access*, vol. 8, pp. 4806–4813, 2020.

[42] B. Kang, S. Xie, M. Rohrbach, Z. Yan, A. Gordo, J. Feng, and Y. Kalantidis, "Decoupling representation and classifier for long-tailed recognition," 2019, *arXiv:1910.09217*.

[43] K. Cao, C. Wei, A. Gaidon, N. Arechiga, and T. Ma, "Learning imbalanced datasets with label-distribution-aware margin loss," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2019, pp. 1–12.

[44] J. Wu, L. Song, T. Wang, Q. Zhang, and J. Yuan, "Forest R-CNN: Large-vocabulary long-tailed object detection and instance segmentation," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 1570–1578.

[45] A. Gupta, P. Dollár, and R. Girshick, "LVIS: A dataset for large vocabulary instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5351–5359.

[46] Y. Li, T. Wang, B. Kang, S. Tang, C. Wang, J. Li, and J. Feng, "Overcoming classifier imbalance for long-tail object detection with balanced group softmax," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10988–10997.

[47] J. Ren, C. Yu, S. Sheng, X. Ma, H. Zhao, S. Yi, and H. Li, "Balanced meta-softmax for long-tailed visual recognition," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2020, pp. 4175–4186.

[48] J. Tan, X. Lu, G. Zhang, C. Yin, and Q. Li, "Equalization loss v2: A new gradient balance approach for long-tailed object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1685–1694.

[49] S. Zhang, Z. Li, S. Yan, X. He, and J. Sun, "Distribution alignment: A unified framework for long-tail visual recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2361–2370.

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